



Marketing Mix Modeling using Google Meridian

MARKETING ANALYTICS LABORATORY
PROJECT

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BUSINESS PROBLEM

Companies invest significant amounts in digital marketing, but understanding which channels actually drive customer purchases remains challenging.



PROJECT OBJECTIVES

- ✓ EVALUATE MARKETING EFFECTIVENESS
- ✓ MEASURE ROI AND MROI

- ✓ EXPLAIN MODEL PREDICTIONS
- ✓ OPTIMIZE BUDGET ALLOCATION

DATASET

Target Variable

ALL_PURCHASES

Media Channels

- Google Paid Search
- Google Shopping
- Google Performance Max
- Google Display

Controls

- Year
- Month

Geo

Organisation ID

Raw Dataset



Cleaning



Aggregation

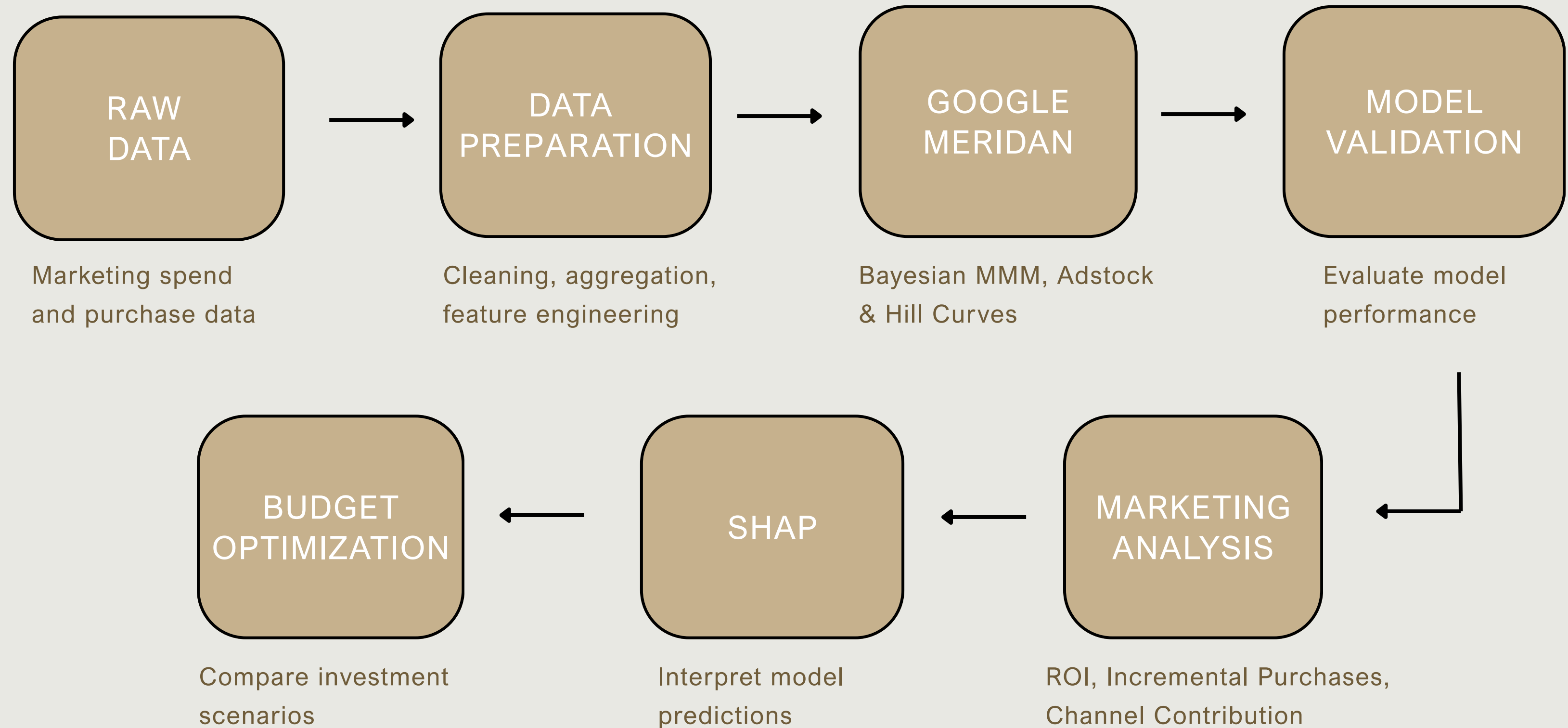


Feature Engineering



Final Dataset

METHODOLOGY



MODEL EVALUATION

Metric	Value
Health Score	81.3
R ²	0.4646
MAPE	0.3295
Weighted MAPE	0.2478

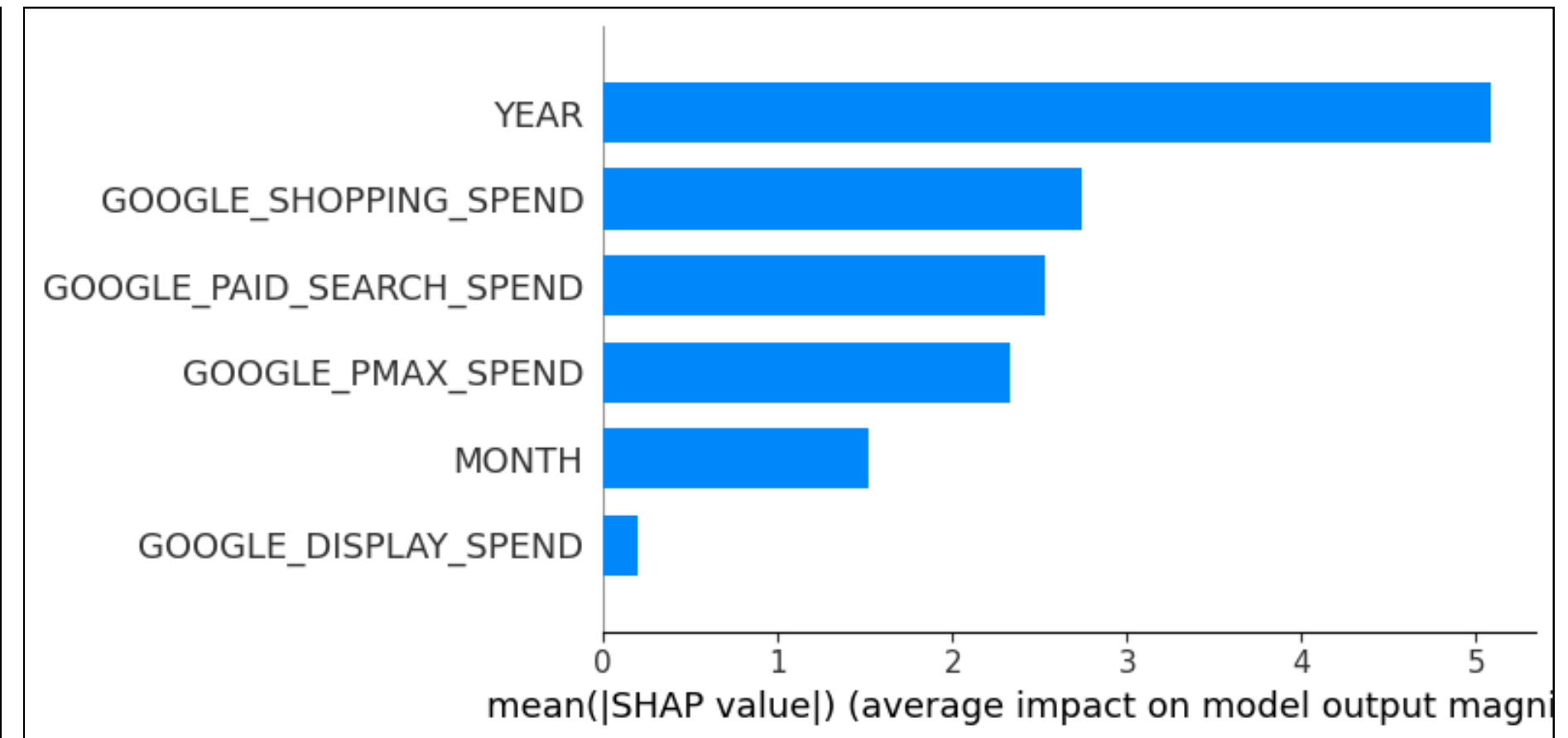
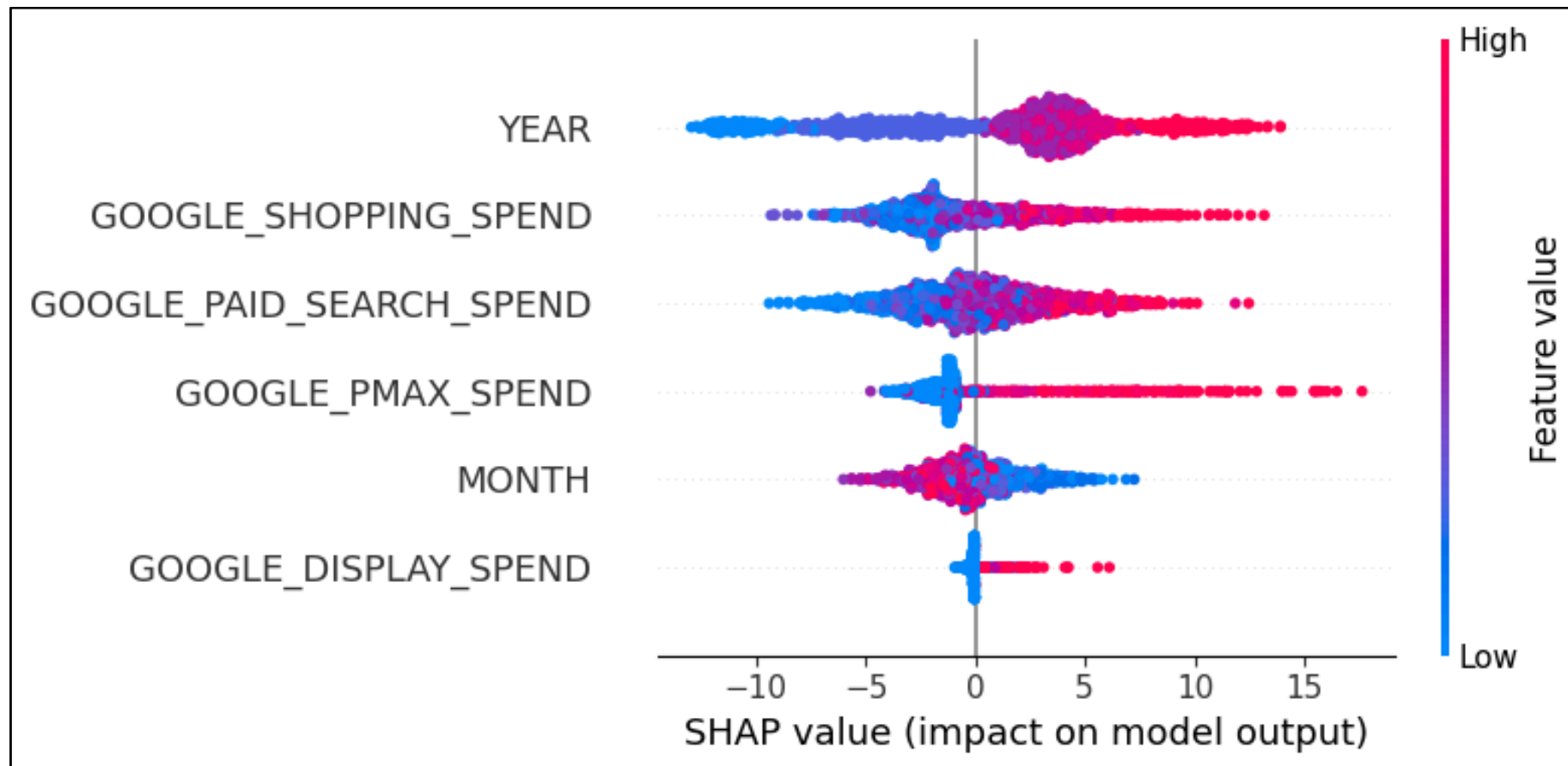
The model achieved stable convergence and acceptable predictive performance, providing a reliable basis for marketing effectiveness analysis and budget optimization.

MARKETING EFFECTIVENESS ANALYSIS

	Channel	Incremental Purchases
0	Google Paid Search	35010.949219
1	Google Shopping	4920.808105
2	Google PMAX	2446.907227
3	Google Display	35.150902

- Key Insights
- Google Paid Search generated the highest incremental purchases.
 - Google Shopping contributed meaningfully but at a lower level.
 - PMAX had a moderate impact.
 - Google Display contributed very little.

SHAP EXPLAINABILITY



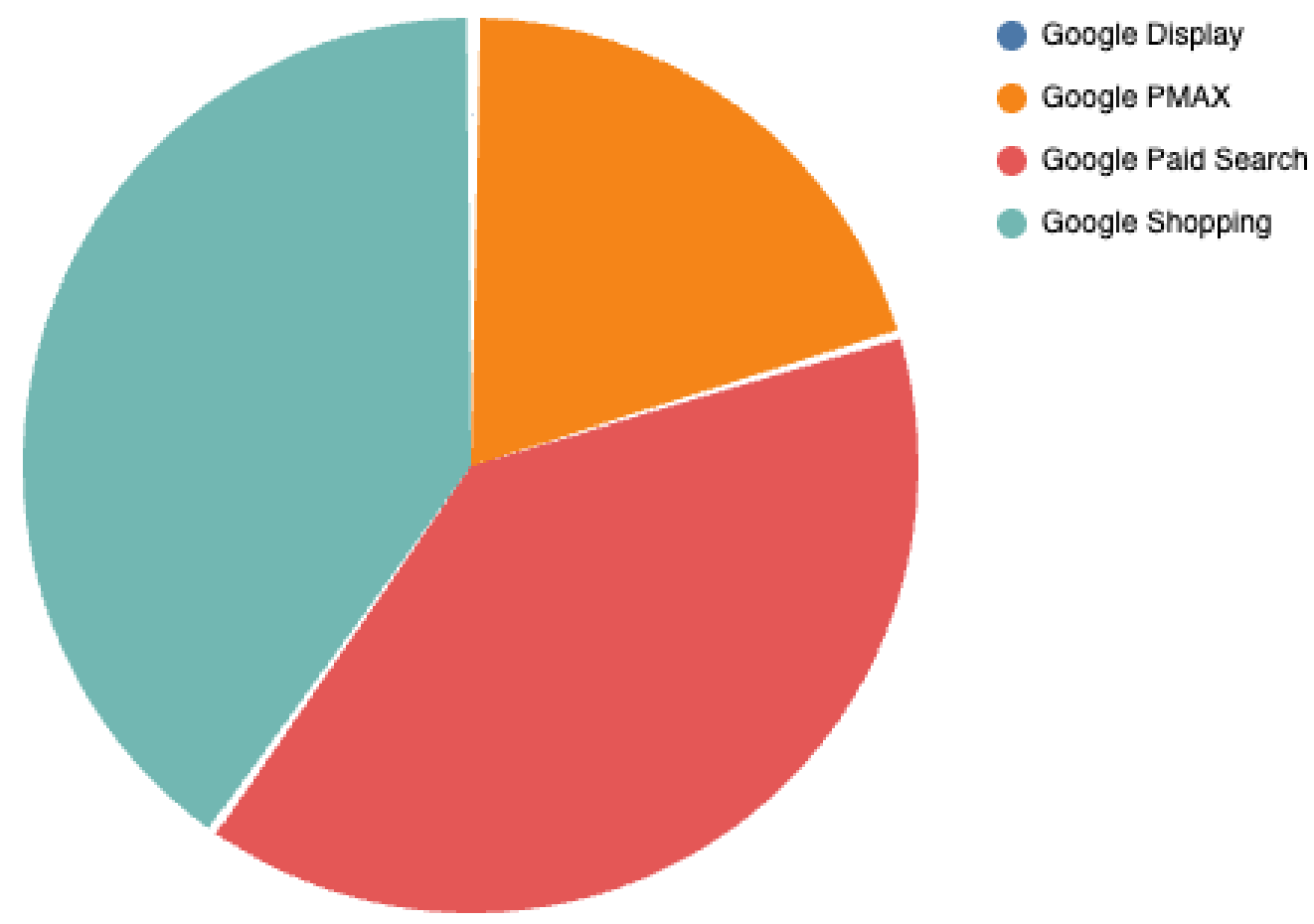
Key Findings

- Marketing spend variables had the greatest impact.
- Seasonal variables also influenced predictions.
- SHAP improved the transparency of the model.

BUDGET OPTIMIZATION

	Scenario	Budget	Paid Search (%)	Shopping (%)	PMAX (%)	Display (%)	Predicted Purchases	Average ROI	Average mROI
0	Scenario 1 (+20%)	588800.0	39.52	40.20	20.11	0.17	49176.48	0.063	0.024
1	Scenario 2 (+50% Paid Search)	490700.0	51.48	30.94	17.44	0.14	48523.58	0.065	0.026
2	Scenario 3 (-30% Saturated)	490700.0	43.29	40.43	16.14	0.14	46777.51	0.069	0.028
3	Scenario 4 (+10% Random)	490700.0	42.18	38.13	19.50	0.18	46475.62	0.068	0.028

Optimized budget allocation



Recommended Strategy

- Increase investment in Google Paid Search.
- Maintain investment in Google Shopping.
- Limit further investment in PMAX and Display.
- Reallocate budget rather than increasing spend uniformly.

BUSINESS RECOMMENDATIONS

INCREASE
INVESTMENT IN
HIGH-
PERFORMING
CHANNELS

REDUCE SPENDING
ON SATURATED
CHANNELS

USE
OPTIMIZATION
BEFORE
CAMPAIGNS

USE SHAP TO
IMPROVE
TRANSPARENCY

CONTINUOUSLY
MONITOR
MARKETING
PERFORMANCE

FINAL RECOMMENDATION

Recommended Strategy for the Next Quarter

Objective: Maximize revenue.

Recommendation:

- Increase the overall marketing budget by 20% (Scenario 1).
- Prioritize additional investment in Google Paid Search.
- Maintain investment in Google Shopping.
- Limit further spending on PMAX and Google Display, where returns begin to diminish.
- Continue using Meridian to monitor performance and adjust future budget allocations.

Expected Outcome: Highest predicted purchases (49,176) while maintaining a strong return on investment.

CONCLUSION

Key Takeaways

- Successfully developed a Marketing Mix Model using Google Meridian.
- Evaluated the effectiveness of four Google advertising channels.
- Found that Google Paid Search was the strongest contributor to incremental purchases.
- Demonstrated how budget optimization can improve marketing performance and support better investment decisions.
- Showed that data-driven budget allocation is more effective than simply increasing marketing spend.